Introduction to optimal transport

The history of optimal transport started long-time ago, in France, few years before the revolution, when Gaspard Monge proposed the following problem in a report that he submitted to the *Académie des Sciences* ([239])[3]. Given two densities of mass $f, g \geq 0$ on $\mathbb{R}^d$, with $\int f(x)\,\mathrm{d}x = \int g(y)\,\mathrm{d}y = 1$, find a map $\mathrm{T}: \mathbb{R}^d \to \mathbb{R}^d$ pushing the first one onto the other, i.e. such that

$$\int_A g(y)\,\mathrm{d}y = \int_{\mathrm{T}^{-1}(A)} f(x)\,\mathrm{d}x \quad \text{for any Borel subset } A \subset \mathbb{R}^d \tag{1}$$

and minimizing the quantity

$$M(\mathrm{T}) := \int_{\mathbb{R}^d} |\mathrm{T}(x) - x| f(x)\,\mathrm{d}x$$

among all the maps satisfying this condition. This means that we have a collection of particles, distributed according to the density f on $\mathbb{R}^d$, that have to be moved so that they form a new distribution whose density is prescribed and is g. The movement has to be chosen so as to minimize the average displacement. In the description of Monge, the starting density f represented a distribution of sand, that had to be moved to a target configuration g. These two configurations correspond to what was called in French *déblais* and *remblais*. Obviously the dimension of the space was only supposed to be $d = 2$ or 3. The map T describes the movement (that we must choose in an optimal way), and $\mathrm{T}(x)$ represents the destination of the particle originally located at x.

In the following, we will often use the image measure of a measure μ on X (measures will indeed replace the densities f and g in the most general formulation of the problem) through a measurable map $\mathrm{T}: X \to Y$: it is the measure on Y denoted by $\mathrm{T}_\#\mu$ and characterized, as in (1), by

$$(\mathrm{T}_\#\mu)(A) = \mu(\mathrm{T}^{-1}(A)) \quad \text{for every measurable set } A,$$
$$\text{or } \int_Y \phi \,\mathrm{d}(\mathrm{T}_\#\mu) = \int_X (\phi \circ \mathrm{T})\,\mathrm{d}\mu \quad \text{for every measurable function } \phi.$$

[3] This happened in 1781, but we translate his problem into modern mathematical language.

More generally, we can consider the problem

$$\text{(MP)} \qquad \min\{M(\mathrm{T}) := \int c(x,\mathrm{T}(x))\,\mathrm{d}\mu(x) \;:\; \mathrm{T}_\#\mu = \nu\},$$

for a more general transport cost $c : X \times Y \to \mathbb{R}$.

When we stay in the Euclidean setting, with two measures μ,ν induced by densities f,g, it is easy – just by a change-of-variables formula – to transform the equality $\nu = \mathrm{T}_\#\mu$ into the PDE

$$g(\mathrm{T}(x))\det(D\mathrm{T}(x)) = f(x), \tag{2}$$

if we suppose f,g and T to be regular enough and T to be injective.

Yet, this equation is highly nonlinear in T and this is one of the difficulties preventing from an easy analysis of the Monge Problem. For instance: how to prove existence of a minimizer? Usually what one does is the following: take a minimizing sequence T_n, find a bound on it giving compactness in some topology (here, if the support of ν is compact the maps T_n take value in a common bounded set, $\mathrm{spt}(\nu)$, and so one can get compactness of T_n in the weak-* L^∞ convergence), take a limit $\mathrm{T}_n \rightharpoonup \mathrm{T}$ and prove that T is a minimizer. This requires semicontinuity of the functional M with respect to this convergence (which is true in many cases, for instance if c is convex in its second variable) for this convergence: we need $\mathrm{T}_n \rightharpoonup \mathrm{T} \Rightarrow \liminf_n M(\mathrm{T}_n) \geq M(\mathrm{T})$), but we also need that the limit T still satisfies the constraint. Yet, the nonlinearity of the PDE prevents from proving this stability when we only have weak convergence (the reader can find an example of a weakly converging sequence such that the corresponding image measures do not converge as an exercise; it is actually **Ex**(1) in the list of exercises).

In [239] Monge analyzed fine questions on the geometric properties of the solution to this problem and he underlined several important ideas that we will see in Chapter 3: the fact that transport rays do not meet, that they are orthogonal to a particular family of surfaces, and that a natural choice along transport rays is to order the points in a monotone way. Yet, he did not really solve the problem. The question of the existence of a minimizer was not even addressed. In the following 150 years, the optimal transport problem mainly remained intimately French, and the *Académie des Sciences* offered a prize on this question. The first prize was won by P. Appell [22] with a long mémoire which improved some points, but was far from being satisfactory (and did not address neither the existence issue[4]).

The problem of Monge has stayed with no solution (does a minimizer exist? how to characterize it?. . .) till progress was made in the 1940s. Indeed, only with the work by Kantorovich (1942, see [201]) it was inserted into a suitable framework which gave the possibility to attack it and, later, to provide solutions and study them. The problem has then been widely generalized, with very general cost functions $c(x,y)$ instead of the Euclidean distance $|x-y|$ and more general measures and spaces. The main idea by Kantorovich is that of looking at Monge's problem as connected to linear programming. Kantorovich indeed decided to change the point of view, and to describe the movement of the particles via a measure γ on $X \times Y$, satisfying $(\pi_x)_\#\gamma = \mu$ and $(\pi_y)_\#\gamma = \nu$. These probability measures over $X \times Y$ are an alternative way to describe

[4]The reader can see [182] - in French, sorry - for more details on these historical questions about the work by Monge and the content of the papers presented for this prize.

the displacement of the particles of μ: instead of giving, for each x, the destination $\mathrm{T}(x)$ of the particle originally located at x, we give for each pair (x,y) the number of particles going from x to y. It is clear that this description allows for more general movements, since from a single point x particles can a priori move to different destinations y. If multiple destinations really occur, then this movement cannot be described through a map T. The cost to be minimized becomes simply $\int_{X\times Y} c\,\mathrm{d}\gamma$. We have now a linear problem, under linear constraints. It is possible to prove existence of a solution, but also to use techniques from convex optimization, such as *duality*, in order to characterize the optimal solution (see Chapter 1).

In some cases, and in particular if $c(x,y) = |x-y|^2$ (another very natural cost, with many applications in physical modeling because of its connection with kinetic energy), it is even possible to prove that the optimal γ does not allow this splitting of masses. Particles at x are only sent to a unique destination $\mathrm{T}(x)$, thus providing a solution to the original problem by Monge. This is what is done by Brenier in [83], where he also proves a very special form for the optimal map: the optimal T is of the form $\mathrm{T}(x) = \nabla u(x)$, for a convex function u. This makes, by the way, a strong connection with the Monge-Ampère equation. Indeed, from (2) we get

$$\det(D^2 u(x)) = \frac{f(x)}{g(\nabla u(x))},$$

which is a (degenerate and non-linear) elliptic equation exactly in the class of convex functions. Brenier also uses this result to provide an original *polar factorization* theorem for vector maps (see Section 1.7.2): vector fields can be written as the composition of the gradient of a convex function and of a measure-preserving map. This generalizes the fact that matrices can be written as the product of a symmetric positive-definite matrix and an orthogonal one.

The results by Brenier can be easily adapted to other costs, strictly convex functions of the difference $x-y$. They have also been adapted to the squared distance on Riemannian manifolds (see [231]). But the original cost proposed by Monge, the distance itself, was more involved.

After the French school, the time of the Russians arrived. From the precise approach introduced by Kantorovich, Sudakov ([290]) proposed a solution for the original Monge problem (MP). The optimal transport plan γ in the Kantorovich problem with cost $|x-y|$ has the following property: the space $\mathbb{R}^d$ can be decomposed in an essentially disjoint union of segments that are preserved by γ (i.e. γ is concentrated on pairs (x,y) belonging to the same segment). This segments are built from a Lipschitz function, whose level set are the surfaces "foreseen" by Monge. Then, it is enough to reduce the problem to a family of 1D problems. If $\mu \ll \mathscr{L}^d$, the measures that μ induces on each segment should also be absolutely continuous, and have no atoms. And in dimension one, as soon as the source measure has no atoms, then one can define a monotone increasing transport, which is optimal for any convex cost of the difference $x-y$.

The strategy is clear, but there is a small drawback: the absolute continuity of the disintegrations of μ along segments, which sounds like a Fubini-type theorem, fails for arbitrary families of segments. Some regularity on their directions is needed. This has been observed by Ambrosio, and fixed in [8, 11]. In the meanwhile, other proofs were obtained by Evans-Gangbo, [160] (via a method which is linked to what we will see in Chapter 4, and unfortunately under stronger assumptions on the densities) and by Caffarelli-Feldman-McCann, [102] and, independently, Trudinger-Wang, [291], via an approximation through strictly convex costs.

After many efforts on the existence of an optimal map, its regularity properties have also been studied: the main reference in this framework is Caffarelli, who proved regularity in the quadratic case, thanks to a study of the Monge-Ampère equation above. Surprisingly, at the beginning of the present century Ma-Trudinger-Wang ([220]) found out the key for the regularity under different costs. In particular they found a condition on costs $c \in C^4$ on $\mathbb{R}^d$ (some inequalities on their fourth-order derivatives) which ensured regularity. It can be adapted to the case of squared distances on smooth manifolds, where the assumption becomes a condition on the curvature of the underlying manifolds. These conditions have later been proven to be sharp by Loeper in [217]. Regularity is a beautiful and delicate matter, which cannot have in this book all the attention that it would deserve. We refer anyway to Section 1.7.6 for more details and references.

But the theory of optimal transport cannot be reduced to the existence and the properties of optimal maps. The success of this theory can be associated to the many connections it has with many other branches of mathematics. Some of these connections pass through the use of the optimal map, yes: think at some geometric and functional inequalities that can be proven (or reproven) in this way. In this book we only present the isoperimetric inequality (Section 2.5.3) and the Brunn-Minkowski one (Section 7.4.2). We stress that one of the most refined advances in quantitative isoperimetric inequalities is a result by Figalli-Maggi-Pratelli, strongly using optimal transport as a tool for their proof, [169].

On the other hand, many applications of optimal transport pass, instead, through the distances it defines (Wasserstein distances, see Chapter 5). Indeed, it is possible to associate to every pair of measure a quantity, denoted by $W_p(\mu,\nu)$, based on the minimal cost to transport μ onto ν for the cost $|x-y|^p$. It can be proven to be a distance, and to metrize the weak convergence of probability measures (at least on compact spaces). This distance is very useful in the study of some PDEs. Some evolution equations, in particular of parabolic type, possibly with non-local terms, can be interpreted as gradient flows (curves of steepest descent, see Chapter 8) for this distance. This idea has first been underlined in [199, 246]. It can be used to provide existence results or approximation of the solutions. For other PDEs, the Wasserstein distance, differentiated along two solutions, may be a technical tool to give stability and uniqueness results, or rate of convergence to a steady state (see Section 5.3.5). Finally, other evolution models are connected either to the minimization of some actions involving the kinetic energy, as it is standard in physics (the speed of a curve of densities computed w.r.t. the W_2 distance is exactly a form of kinetic energy), or to the gradient of some special convex functions appearing in the PDE (see Section 8.4.4).

The structure of the space $\mathbb{W}_p$ of probability measures endowed with the distance W_p also received and still receives a lot of attention. In particular, the study of its geodesics and of the convexity properties of some functionals along these geodesics have been important, both because they play a crucial role in the metric theory of gradient flows developed in the reference book [17] and because of their geometrical consequences. The fact that the entropy $E(\rho) := \int \rho \log \rho$ is or not convex along these geodesics turned out to be equivalent, on Riemannian manifolds, to lower bounds on the Ricci curvature of the manifold. This gave rise to a wide theory of analysis in metric measure spaces, where this convexity property was chosen as a definition for the curvature bounds (see [219, 288, 289]). This theory underwent many progresses in these last years, thanks to the many recent results by Ambrosio, Gigli, Savaré, Kuwada, Ohta and their collaborators (see, as an example, [184, 18]).

From the point of view of modeling, optimal transport may appear in many fields more or less linked to spatial economics, traffic, networks and collective motions, but the pertinence of the Monge-Kantorovich model can be questioned, at least for some specific applications. This leads to the study of alternative model, either more "convex" (for traffic congestion), or more "concave" (for the organization of an efficient transport network). These models are typically built under the form of a flow minimization under divergence constraints, and are somehow a variant of the original Monge cost. Indeed, the original Monge problem (optimal transport from μ to ν with cost $|x-y|$) is also equivalent to the problem of minimizing the L^1 norm of a vector field $\mathbf{w}$ under the condition $\nabla \cdot \mathbf{w} = \mu - \nu$. This very last problem is presented in Chapter 4, and the traffic congestion and branched transport models are presented as a variant in Section 4.4.

Finally, in particular due to its applications in image processing (see Section 2.5.1 and 5.5.5), it has recently become crucial to have efficient ways of computing, or approximating, the optimal transport or the Wasserstein distances between two measures. This is a new and very lively field of investigation: the methods that are presented in Chapter 6 are only some classical ones. This book does not want to be exhaustive on this point, but one of its claimed goals is for sure to enhance the interest on these subjects.

It is not our intention to build a separating wall between two sides of optimal transportation, the "pure" one, and the "applied one". Both are progressing at an impressive rate in these last years. This book is devoted to those topics in the theory that could be more interesting for the reader who looks at modeling issues in economics, image processing, social and biological evolutionary phenomena, and fluid mechanics, at the applications to PDEs, and at numerical issues. It would be impossible to summarize the new directions that these topics are exploring in this short introduction, and also the book cannot do it completely.

We will only try to give a flavor of these topics, as well as a rigorous analysis of the mathematics which are behind them.